

# bge-m3与bge-reranker模型介绍\*

## 课程目标

- 知道bge-m3模型是如何得到稀疏向量和稠密向量
- 知道bge-reranker模型的使用

## bge-m3模型

论文地址:

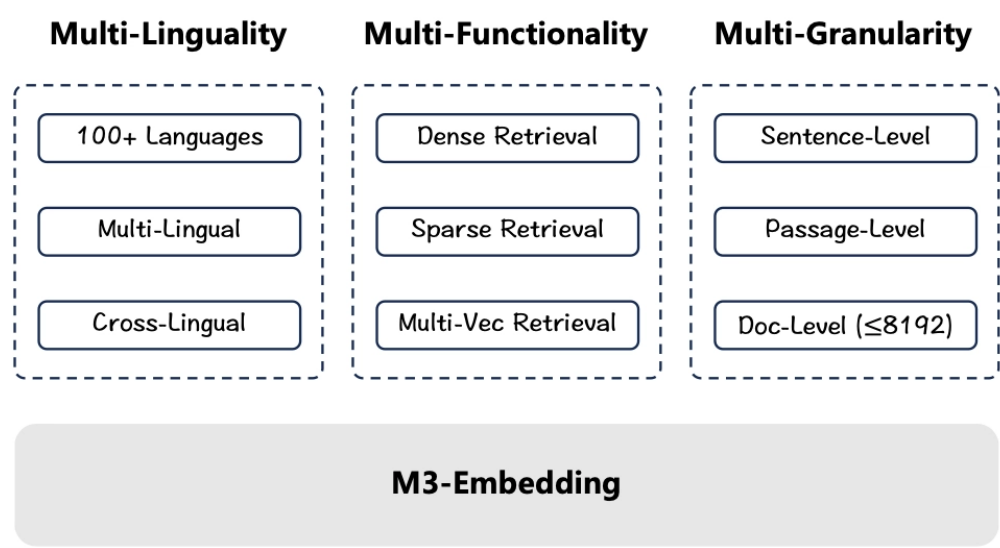


Figure 1: Characters of M3-Embedding.

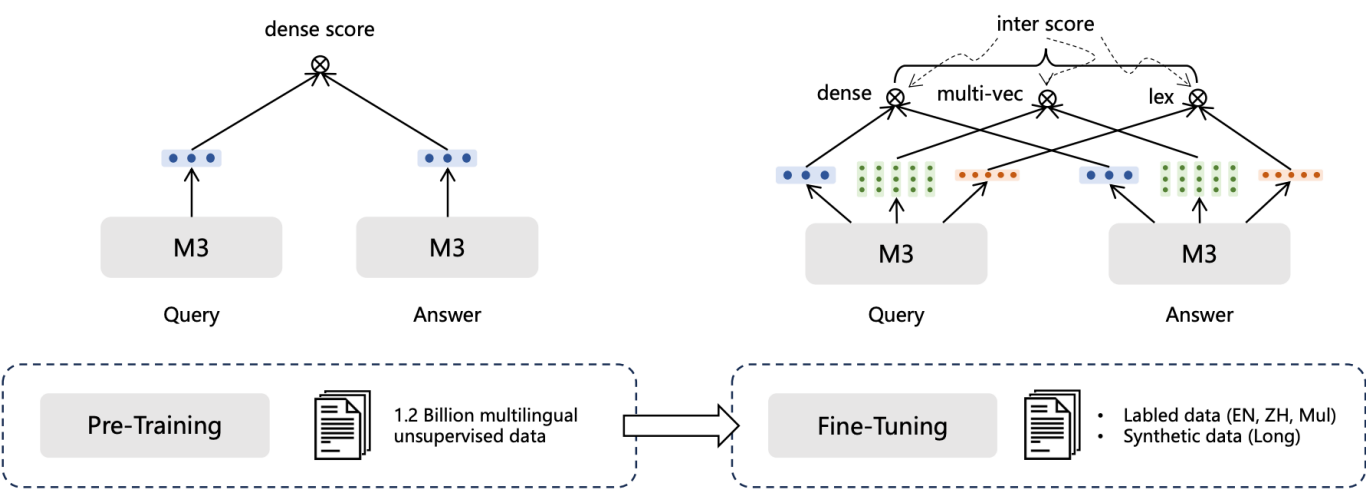


Figure 2: Multi-stage training process of M3-Embedding with self-knowledge distillation.

使用bge-m3模型对文本进行编码，可以得到稀疏向量和稠密向量，用于后续的混合检索，实际上bge-m3模型还能编码多维向量，实现多向量的检索。下面分别介绍这几种检索方式。

|       | Baselines (Prior Work) |      |             |                      |                          |          | M3-Embedding (Our Work) |        |             |              |             |
|-------|------------------------|------|-------------|----------------------|--------------------------|----------|-------------------------|--------|-------------|--------------|-------------|
|       | BM25                   | mDPR | mContriever | mE5 <sub>large</sub> | E5 <sub>mistral-7b</sub> | OpenAI-3 | Dense                   | Sparse | Multi-vec   | Dense+Sparse | All         |
| ar    | 18.9                   | 48.2 | 58.2        | 68.7                 | 59.6                     | 65.6     | 71.1                    | 23.5   | 71.4        | 71.1         | <b>71.5</b> |
| da    | 49.3                   | 67.4 | 73.9        | 77.4                 | <b>77.8</b>              | 73.6     | 77.2                    | 55.4   | 77.5        | 77.4         | 77.6        |
| de    | 35.4                   | 65.8 | 71.7        | 76.9                 | <b>77.0</b>              | 73.6     | 76.2                    | 43.3   | 76.3        | 76.4         | 76.3        |
| es    | 43.4                   | 66.8 | 72.6        | 76.4                 | <b>77.4</b>              | 73.9     | 76.4                    | 50.6   | 76.6        | 76.7         | 76.9        |
| fi    | 46.3                   | 56.2 | 70.2        | 74.0                 | 72.0                     | 72.7     | 75.1                    | 51.1   | 75.3        | 75.3         | <b>75.5</b> |
| fr    | 45.3                   | 68.2 | 72.8        | 75.5                 | <b>78.0</b>              | 74.1     | 76.2                    | 53.9   | 76.4        | 76.6         | 76.6        |
| he    | 26.9                   | 49.7 | 63.8        | 69.6                 | 47.2                     | 58.1     | 72.4                    | 31.1   | 72.9        | 72.5         | <b>73.0</b> |
| hu    | 38.2                   | 60.4 | 69.7        | 74.7                 | <b>75.0</b>              | 71.2     | 74.7                    | 44.6   | 74.6        | 74.9         | <b>75.0</b> |
| it    | 45.2                   | 66.0 | 72.3        | 76.8                 | <b>77.1</b>              | 73.6     | 76.0                    | 52.5   | 76.4        | 76.3         | 76.5        |
| ja    | 24.5                   | 60.3 | 64.8        | 71.5                 | 65.1                     | 71.9     | 75.0                    | 31.3   | 75.1        | 75.0         | <b>75.2</b> |
| km    | 27.8                   | 29.5 | 26.8        | 28.1                 | 34.3                     | 33.9     | 68.6                    | 30.1   | 69.1        | 68.8         | <b>69.2</b> |
| ko    | 27.9                   | 50.9 | 59.7        | 68.1                 | 59.4                     | 63.9     | 71.6                    | 31.4   | 71.7        | 71.6         | <b>71.8</b> |
| ms    | 55.9                   | 65.5 | 74.1        | 76.3                 | 77.2                     | 73.3     | 77.2                    | 62.4   | <b>77.4</b> | <b>77.4</b>  | <b>77.4</b> |
| nl    | 56.2                   | 68.2 | 73.7        | 77.8                 | <b>79.1</b>              | 74.2     | 77.4                    | 62.4   | 77.6        | 77.7         | 77.6        |
| no    | 52.1                   | 66.7 | 73.5        | 77.3                 | 76.6                     | 73.3     | 77.1                    | 57.9   | 77.2        | <b>77.4</b>  | 77.3        |
| pl    | 40.8                   | 63.3 | 71.6        | 76.7                 | <b>77.1</b>              | 72.7     | 76.3                    | 46.1   | 76.5        | 76.3         | 76.6        |
| pt    | 44.9                   | 65.5 | 72.0        | 73.5                 | <b>77.5</b>              | 73.7     | 76.3                    | 50.9   | 76.4        | 76.5         | 76.4        |
| ru    | 33.2                   | 62.7 | 69.8        | <b>76.8</b>          | 75.5                     | 72.0     | 76.2                    | 36.9   | 76.4        | 76.2         | 76.5        |
| sv    | 54.6                   | 66.9 | 73.2        | 77.6                 | <b>78.3</b>              | 74.0     | 76.9                    | 59.6   | 77.2        | 77.4         | 77.4        |
| th    | 37.8                   | 53.8 | 66.9        | 76.0                 | 67.4                     | 65.2     | 76.4                    | 42.0   | 76.5        | 76.5         | <b>76.6</b> |
| tr    | 45.8                   | 59.1 | 71.1        | 74.3                 | 73.0                     | 71.8     | 75.6                    | 51.8   | 75.9        | <b>76.0</b>  | <b>76.0</b> |
| vi    | 46.6                   | 63.4 | 70.9        | 75.4                 | 70.9                     | 71.1     | 76.6                    | 51.8   | 76.7        | 76.8         | <b>76.9</b> |
| zh_cn | 31.0                   | 63.7 | 68.1        | 56.6                 | 69.3                     | 70.7     | 74.6                    | 35.4   | 74.9        | 74.7         | <b>75.0</b> |
| zh_hk | 35.0                   | 62.8 | 68.0        | 58.1                 | 65.1                     | 69.6     | 73.8                    | 39.8   | 74.1        | 74.0         | <b>74.3</b> |
| zh_tw | 33.5                   | 64.0 | 67.9        | 58.1                 | 65.8                     | 69.7     | 73.5                    | 37.7   | 73.5        | <b>73.6</b>  | <b>73.6</b> |
| Avg   | 39.9                   | 60.6 | 67.9        | 70.9                 | 70.1                     | 69.5     | 75.1                    | 45.3   | 75.3        | 75.3         | <b>75.5</b> |

Table 2: Cross-lingual retrieval performance on MKQA (measured by Recall@100).

稠密向量检索

稠密向量是指，维度较低每个位置的上基本都有数值，通常也叫做**低维稠密向量**。bge-m3模型编码编码以后默认长度为1024维。

输入query编码以后的向量为  $H_q$ ，使用[CLS]输出的向量作为全局向量来代表query得语义，于是有：

$$e_q = norm(H_q[0])$$

文档向量为  $e_p$ ，于是就可以计算这两个向量的距离：

$$s_{dense} = f_{sim}(e_q, e_p)$$

其中  $f_{sim}$  距离可以是余弦距离，内积，L2等

代码：

代码块

```
1 from milvus_model.hybrid import BGEM3EmbeddingFunction
2 m3_path = "../rag_qa/models/bge-m3"
3 embedding_function = BGEM3EmbeddingFunction(model_name_or_path=m3_path,
use_fp16=False, device='cpu')
```

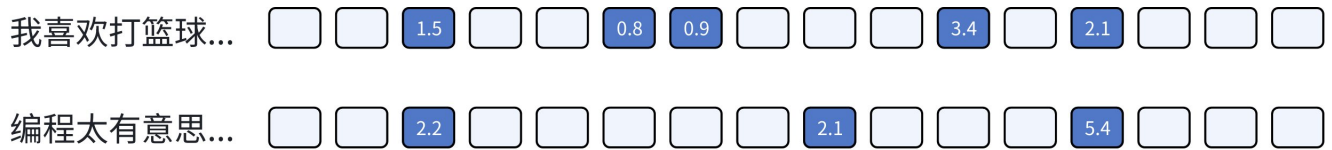
```

4 embed = embedding_function(["黑马JAVA", "黑马大模型"])
5 print(embed["dense"])
6 # [array([-0.03068057,  0.00156495, -0.04899885, ...,  0.04231707,
7 #          0.01612839,  0.0713326 ], shape=(1024,), dtype=float32),
8 # array([-0.01287245, -0.01234254, -0.06556936, ...,  0.04092151,
9 #          -0.01335396,  0.05632111], shape=(1024,), dtype=float32)]

```

## 稀疏向量检索

稀疏向量是指维度很高，大多数维度上的数值为0，只保留非零的数值和位置。



稀疏向量的表示

输入query编码以后的向量为  $H_q$ ，针对每个token  $i$  需要计算对应的重要性，与bm25类似

$$w_{qt} = \text{Relu}(W_{lex}^T H_q[i])$$

其中  $W_{lex} \in \mathbb{R}^{d \times 1}$ ，计算出的  $w_{qt}$  是一个数值，代表token i的重要程度。

于是针对输入p来说，就可以计算query与p的稀疏向量的距离：

$$s_{lex} = \sum_{t \in q \cap p} (w_{qt} * w_{pt})$$

代码：

代码块

```

1 from milvus_model.hybrid import BGEM3EmbeddingFunction
2 m3_path = "../rag_qa/models/bge-m3"
3 embedding_function = BGEM3EmbeddingFunction(model_name_or_path=m3_path,
4 use_fp16=False, device='cpu')
5 embed = embedding_function(["黑马JAVA", "黑马大模型"])
6 row = embed["sparse"][0]
7 print(row.col)
8 # array([ 6, 5958, 7320, 189950])
9 print(row.data)
10 # array([0.00417884, 0.24346864, 0.15046994, 0.28743052])

```

## 多向量检索

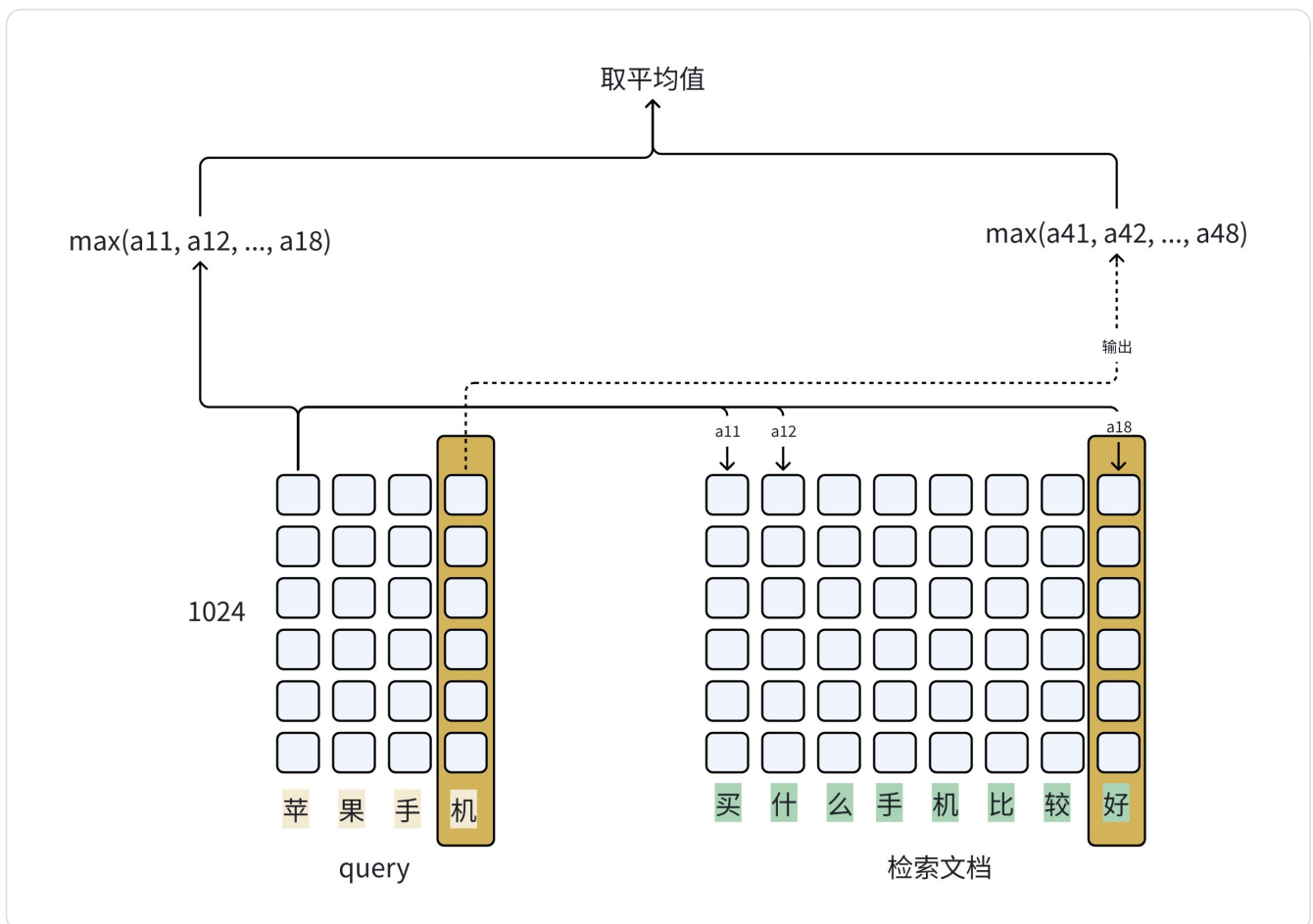
$$E_q = \text{norm}(W_{mul}^T H_q)$$

$$E_p = \text{norm}(W_{mul}^T H_p)$$

其中  $W_{mul}^T \in \mathbb{R}^{d \times d}$ ，是一个可学习的矩阵。

计算两个文档的距离如下：

$$s_{mul} = \frac{1}{N} \sum_{i=1}^N \max_{j=1}^M E_q[i] \cdot E_p^T[j]$$

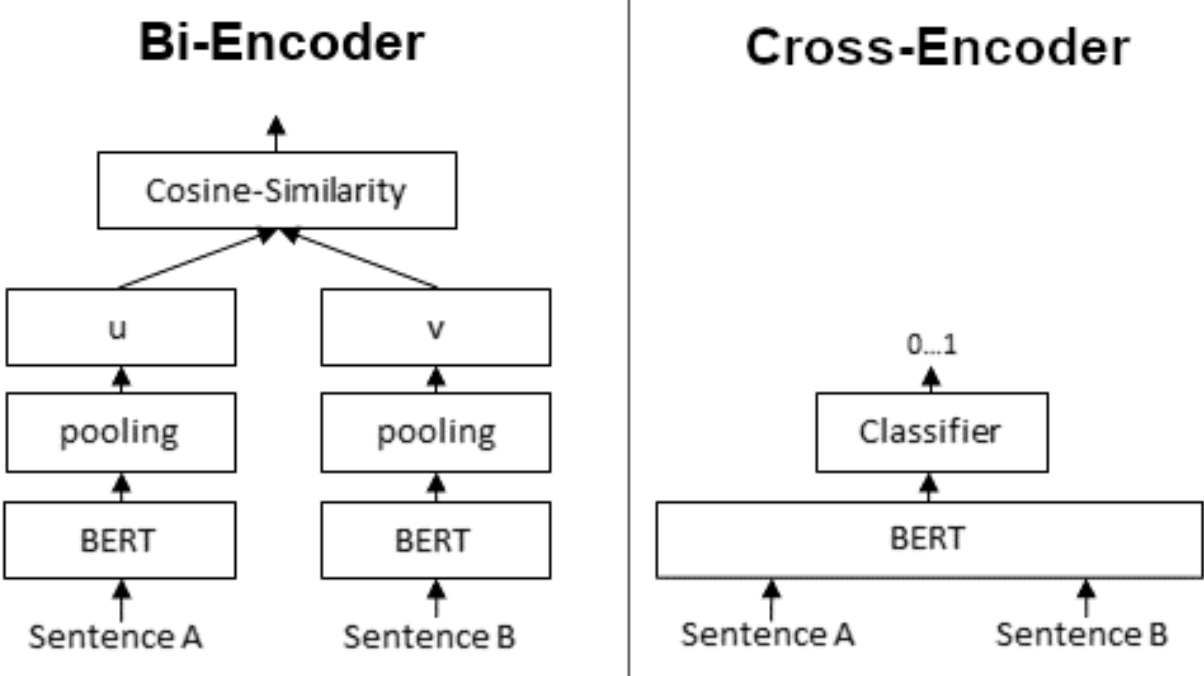


## bge-reranker-large模型

[XLM-RoBERT-Base论文地址](#)

bge-reranker模型包含两个模型分别是base和large，分别是基于 [XLM-RoBERTa-Base](#)和[XLM-RoBERTa-Large](#) 模型

| Model                                   | Language          | Parameters | Model Size | Description                                                    |
|-----------------------------------------|-------------------|------------|------------|----------------------------------------------------------------|
| <a href="#">BAAI/bge-reranker-large</a> | English & Chinese | 560M       | 2.24 GB    | Larger reranker model, easy to deploy with better inference    |
| <a href="#">BAAI/bge-reranker-base</a>  | English & Chinese | 278M       | 1.11 GB    | Lightweight reranker model, easy to deploy with fast inference |



总结

| 对比维度   | BGE-M3               | BGE-Reranker-Large     |
|--------|----------------------|------------------------|
| 模型类型   | Bi-encoder Embedding | Cross-encoder Reranker |
| 原始架构   | XLM-RoBERTa          | XLM-RoBERTa            |
| 主要用途   | 召回 / 向量化             | 精排 / 评分                |
| 训练目标   | InfoNCE 对比损失 + 自蒸馏   | 监督排序损失                 |
| 文本长度支持 | ≈ 8192 tokens        | ≈ 512 tokens (典型)      |
| 多语种能力  | 强 (100+ 语言)          | 中等 (支撑中英)              |
| 优势     | 快速召回，高效检索            | 高精度排序                  |